Cem Ozcelik

+90 553 473 4556 | i.cemozcelik@gmail.com | linkedin.com/in/cemozcelik | github.com/ccemozclk | Istanbul, Turkey

PROFESSIONAL SUMMARY

Senior Data Scientist with an Industrial Engineering background and nearly 5 years of end-to-end ML experience across e-commerce search, recommendation systems, demand forecasting, and MLOps deployment. Built a production microservices semantic search engine on 800K+ product descriptions using SentenceTransformers + Qdrant vector DB with sub-millisecond cached responses (Redis) and dual-engine A/B testing -- the same retrieval foundation that powers RAG and LLM-driven product discovery. Layered ALS collaborative filtering + RFM/K-Means behavioral segmentation on distributed PySpark for personalized recommendations, and Item2Vec embeddings for "frequently bought together." MLOps ownership end-to-end: Docker, FastAPI, Django, model registry, versioned REST. The Industrial Engineering foundation -- operations research, MILP, cost-function optimization -- is the differentiator: I default to "what's the asymmetric cost of being wrong, and what's the latency/quality tradeoff?" rather than chasing accuracy. Demos: github.com/ccemozclk/amazon-ai-search-engine | github.com/ccemozclk/Spark-Recommendation-Engine

CORE COMPETENCIES

Skills: Semantic Search & Retrieval, Vector Databases (Qdrant), Recommendation Systems (ALS, Item2Vec), Customer Segmentation (RFM, K-Means), A/B Testing & Experiment Design, Demand Forecasting, MLOps & Model Lifecycle Management, Deep Learning, Cost-Function Optimization

Tools: Python, scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, SentenceTransformers, PySpark, Databricks, SQL (Advanced), Docker, FastAPI, Django, Streamlit, Redis, Qdrant

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe
Jan 2026 -- Present

Senior Database Analyst | Maltepe, Istanbul

- Scaled a custom Django web application into a **multi-model ML deployment platform** with versioned REST APIs and a model registry pattern, taking new models from notebook to production in days rather than weeks.
- Directed predictive supply chain optimization using Python-based forecasting algorithms, sustaining an **8-hour weekly reduction in operational overtime** and architecting robust pipelines for AI-driven forecasting.
- Translated SKU-level Association Rule Mining outputs into strategic cross-sell actions, driving continuous revenue growth.
- **Tech:** Python, Django, REST APIs, ML pipelines, Association Rule Mining, PostgreSQL, Docker

Anadolu Pet Akvaryum & Effeffe
Sept 2024 -- Dec 2025

IT & Data Analytics Specialist | Maltepe, Istanbul

- Architected a full-stack internal web application from scratch (Python/Django, SOLID principles) -- the deployment foundation later scaled into the ML platform.
- Engineered predictive supply chain models in Python and PL/SQL, **reducing stockouts and improving inventory turnover**.
- Conducted Route-Level Profitability Analysis with advanced SQL, defining cost-to-profit ratios that delivered a **3% improvement in logistics efficiency**.
- **Tech:** Python, Django, PL/SQL, Oracle, SOLID, full-stack deployment

Birinci Automotive
Jan 2022 -- Sept 2024

Mid-Level Data Analyst | Cayirova, Kocaeli

- Built an **unsupervised clustering framework on behavioral time-series data** for anomaly detection -- the same statistical primitives (feature engineering, embedding, threshold optimization) that drive user-behavior modeling and retrieval ranking in e-commerce platforms.
- Delivered a **15% weekly increase in critical machine availability** and **15% OEE improvement** by bridging raw sensor data and operational strategy.
- Engineered ETL pipelines in MSSQL and Python to ingest high-volume manufacturing data, replacing manual reporting workflows.
- **Tech:** Python, MSSQL, ETL, unsupervised clustering, time-series analysis

Birinci Automotive

June 2021 -- Jan 2022

Junior Data Analyst | Cayirova, Kocaeli

- Conducted EDA and applied statistical analysis to workforce datasets, uncovering patterns to optimize HR policies.
- Designed automated data workflows and ingestion pipelines, **improving reporting efficiency by 15%**.
- **Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

E-Commerce AI Search Engine

Semantic Search & Vector DB

Production-ready microservices architecture with **dual-engine A/B testing**: SentenceTransformers (`all-MiniLM-L6-v2`) vs custom-trained Word2Vec on **800K+ product descriptions**. **Qdrant vector DB** for similarity retrieval, Redis for sub-millisecond cached responses, PySpark distributed ETL for massive JSONL ingestion. **Item2Vec collaborative filtering** for "Frequently Bought Together" recommendations on hundreds of thousands of user sessions. Custom LSTM next-token model for intelligent search auto-complete. The retrieval foundation maps directly to RAG-style and LLM-driven product discovery.

PyTorch, FastAPI, SentenceTransformers, Qdrant (Vector DB), Redis, PySpark, Streamlit, Docker | github.com/ccemozclk/amazon-ai-search-engine

Scalable Recommendation Engine

PySpark & Customer Segmentation

End-to-end personalized recommendation system on Apache Spark/Databricks. **RFM analysis + K-Means** for behavioral customer segmentation into 5 strategic tiers (Champions, At-Risk, etc.) -- the same segmentation logic powering e-commerce next-best-offer and CRM personalization. **ALS collaborative filtering** with implicit feedback (purchase quantity proxy) and cold-start handling. Generated logically validated affinities across hundreds of thousands of user sessions at distributed scale.

PySpark, Apache Spark, Databricks, K-Means, ALS, Streamlit | github.com/ccemozclk/Spark-Recommendation-Engine

Retail Demand Forecasting

End-to-End MLOps Pipeline

~12.83% RMSPE on 6-week daily sales across 1,115 Rossmann stores with time-based validation (no leakage). Object-oriented training pipeline (DataIngestion to ModelTrainer). FastAPI REST inference + Streamlit dashboard with dynamic confidence intervals translating ML metrics to business terms. Surfaced actionable insight: B-type stores yield lower promotion ROI vs A/C, enabling marketing budget reallocation -- the kind of operations + analytics overlap that drives e-commerce planning.

Python, LightGBM, XGBoost, Optuna, FastAPI, Streamlit, Docker | github.com/ccemozclk/Rossman-Demand-Forecasting

EDUCATION

B.Sc. Industrial Engineering — Kirikkale University

Sept 2016 -- Feb 2021

Rigorous foundation in statistical analysis, operations research, and optimization. Designed MILP models with IBM ILOG CPLEX. Applied AHP and TOPSIS to multi-criteria decision problems -- direct prerequisite for designing cost/latency/quality tradeoffs in production ML and search systems.

CERTIFICATIONS

Associate Data Scientist Career Track (incl. PySpark, Recommendation Engines) — DataCamp — Jan -- May 2024

Data Science & Machine Learning Bootcamp (6 months, 10+ projects) — Miuul — Mar -- Sept 2022

SKILLS

Languages: Python (Advanced), SQL (Advanced), Bash

ML & DL: scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, h2o.ai, Recommendation Systems (ALS, Item2Vec), Customer Segmentation (RFM, K-Means), Semantic Search, SentenceTransformers, CNNs, RNNs, Transfer Learning, A/B Testing, Statistical Analysis

Big Data: PySpark, Apache Spark, Databricks **MLOps & Deployment:** Docker, FastAPI, Django, Flask, REST APIs, Git, Linux

Databases: PostgreSQL, MSSQL, Oracle, ETL Pipelines, Redis, Qdrant (Vector DB)

Advanced: Cost-Function Optimization, MILP, AHP/TOPSIS, OOP, System Architecture, AWS (EC2, S3)

Languages: Turkish (Native), English (Fluent)